

UNIFIED SPECULATIVE SERVING ENGINE ARCHITECTURE ON TESLA T4

Pre-allocated Static KV-Cache + PromptLookupDraftEngine Eliminating Secondary Model Footprint

PROMPT LOOKUP DRAFT ENGINE

Zero-Weight N-Gram Trie

0 MB Secondary VRAM

Sub-Microsecond Proposal

0.0062 ms proposal time

Candidate Sequence K=3

[tok_1, tok_2, tok_3]

Eliminates 35ms Python Dispatch Tax

TARGET MODEL VERIFICATION

Batched Verification Step

Single Target Forward Pass

Parallel Softmax Match

Exact Greedy Equivalence

Tokens Accepted (M=3)

1.34 - 2.07 tokens / step

Zero Target Model Quantization Drift

STATIC KV CACHE & ROLLBACK

Zero Heap Allocation

Pre-allocated max_seq_len

O(1) Pointer Rollback

0.51 us (14,388x faster)

Net Wall-Clock Speedup

1.48x Net (up to 2.16x)

26.59 → 39.34 tok/s on T4